Brandon Miner

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Position applied for: ______________________________ **Date:** _______________

Introduction

I am a data scientist and machine learning engineer with direct experience building production AI systems in mission-driven, resource-constrained environments. My work sits at the intersection of rigorous technical delivery and direct engagement with the non-technical stakeholders who depend on those systems. At the ACLU of Northern California, I owned the full lifecycle of a production ML inference platform — from eliciting requirements from civil rights investigators, through GPU-accelerated pipeline engineering, to on-premises deployment on client hardware — replacing over 300 hours of manual review per monthly cycle with automated, audit-ready analysis. Prior to that, I led a data science team at the San Diego County Taxpayers Association, turning messy public records into statistical findings presented directly to county officials. I hold an M.S. in Data Science and Artificial Intelligence from the University of San Francisco (2026) and a B.S. in Statistics with Emphasis in Data Science from San Diego State University (2025).

Qualification 1 — Machine Learning Systems Design and Production Deployment

During my tenure at the ACLU of Northern California, I architected and shipped a multi-modal AI platform entirely from scratch. The system ingests batches of 600+ police bodycam videos per month, runs GPU-accelerated transcription via OpenAI Whisper and FFmpeg, and feeds a multi-stage NLP classification pipeline that combines LLM-assisted auto-labeling, iterative fine-tuning, and structured output parsing. The full stack — a private-server FastAPI backend paired with a React.js web frontend — was deployed on the client's own hardware with zero cloud dependency, a hard constraint imposed by the sensitivity of the footage. I made every architectural decision: data schema, model selection, inference API design, and the iterative fine-tuning loop.

This experience demonstrates hands-on ownership of the complete ML development lifecycle in a production setting, including the ability to make pragmatic engineering decisions when external infrastructure is unavailable or inadvisable.

Qualification 2 — Data Engineering and Pipeline Development

I have built and shipped end-to-end data pipelines across multiple domains. At the ACLU, I designed a GPU-accelerated ETL pipeline that normalizes raw video into transcribed, structured records ready for downstream ML inference. At the San Diego County Taxpayers Association, I constructed a workforce analytics pipeline combining iterative web scraping, NLP-based entity standardization, and regression modeling to quantify the relationship between police staffing levels and wage costs — a 19.7% wage increase per 1% staffing reduction — findings subsequently presented to county officials. In academic project work, I engineered a multi-source agricultural ETL pipeline in Python, containerized the full application with Docker Compose, and deployed it to Google Cloud Run as a live interactive Streamlit dashboard.

Across these projects I have applied consistent engineering practices: schema design before ingestion, reproducible containerized environments, automated CI/CD with GitHub Actions, and clear separation between raw ingestion, transformation, and serving layers.

Qualification 3 — Stakeholder Engagement and Communication

A recurring theme across my roles has been embedding directly with non-technical stakeholders to translate domain knowledge into working systems. At the ACLU, my primary collaborators were civil rights investigators and legal staff with no engineering background. I ran requirements-gathering sessions to surface what signals they needed from bodycam footage, prototyped iteratively against their feedback, and ultimately delivered a self-service interface that allows investigators to run model inference without any engineering support. At the San Diego County Taxpayers Association, I managed a four-analyst team and served as the primary point of contact for government stakeholders, presenting data findings in accessible formats to county officials who informed policy decisions.

I am comfortable operating in both directions across the technical/non-technical divide: translating ambiguous operational needs into concrete system requirements, and explaining ML outputs and their limitations to decision-makers who need actionable clarity.

Qualification 4 — Statistical Analysis and Quantitative Research

My undergraduate training in statistics at San Diego State University provided a rigorous foundation in applied quantitative methods, including regression modeling, mixed-effects models, multicollinearity reduction (VIF analysis), PCA-based dimensionality reduction, and ROC-AUC threshold optimization. At the San Diego County Taxpayers Association, I applied mixed-effects regression to regional homelessness expenditure data to isolate high-impact intervention types — such as rapid rehousing — and surface concrete resource allocation recommendations for program leadership. I also performed threshold-optimized classification for credit card fraud detection, tuning the decision boundary to minimize false negatives while holding false positives below 2%, directly modeling a real business constraint.

My graduate coursework at USF deepened this foundation with applied machine learning, deep learning, distributed computing with Apache Spark, and experiment design — enabling me to move fluidly between statistical rigor and scalable production systems.

Qualification 5 — Civic and Public-Interest Technology

I have deliberately oriented my technical work toward public-interest contexts. At the ACLU, I built tools that support civil rights investigators auditing police conduct, with an explicit goal of expanding oversight capacity. I also led a geospatial privacy analysis initiative to expose deanonymization risks in commercially sold location data, using DBSCAN clustering and the Google Reverse Geocoding API to trace identifiable behavioral patterns (home, work, medical facilities) from purportedly anonymized GPS traces, producing evidentiary exhibits for policy advocacy. At the San Diego County Taxpayers Association, my workforce analytics findings were surfaced directly to county officials and contributed to public reporting on civil service compensation dynamics.

I understand that technical work in civic contexts carries heightened responsibilities around data sensitivity, privacy, and the legibility of outputs to non-specialist audiences. These considerations have shaped every design decision I have made in these environments.

Additional Qualifications

- **Languages & Frameworks:** Python (FastAPI, PyTorch, Pandas, Scikit-learn, Transformers), JavaScript (React.js), SQL (PostgreSQL), Bash, R (tidyverse)
- **Infrastructure & MLOps:** Docker, Docker Compose, Google Cloud Run, GitHub Actions CI/CD, MLflow experiment tracking, private-server and client-hardware deployment, GPU-accelerated inference
- **ML Methods:** NLP pipelines (ASR, text classification, NER), LLM-assisted labeling, multi-stage fine-tuning, LightGBM, mixed-effects regression, PCA, DBSCAN, champion/challenger model evaluation
- **Soft Skills:** cross-functional team leadership (4+ person teams), requirements gathering, stakeholder presentation to government and legal audiences

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I welcome the opportunity to discuss how my background aligns with the specific needs of this role. Thank you for your consideration.

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